

Migration on a network

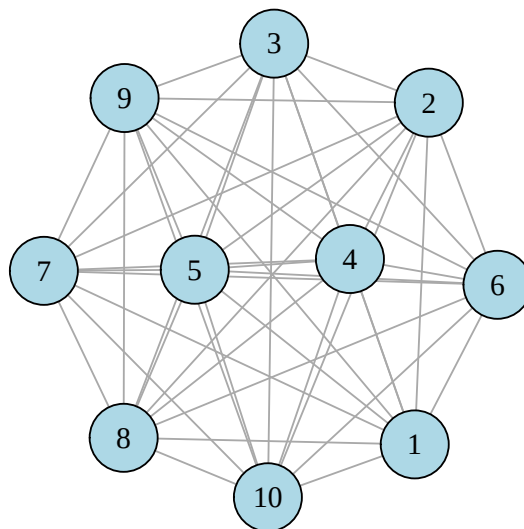
Introduction

Tigers have declined in numbers partly due to loss of habitat connectivity. Here, we use movement along a dynamic network to assess what habitat and connectivity characteristics facilitates or minimizes local extinctions.

Simple network

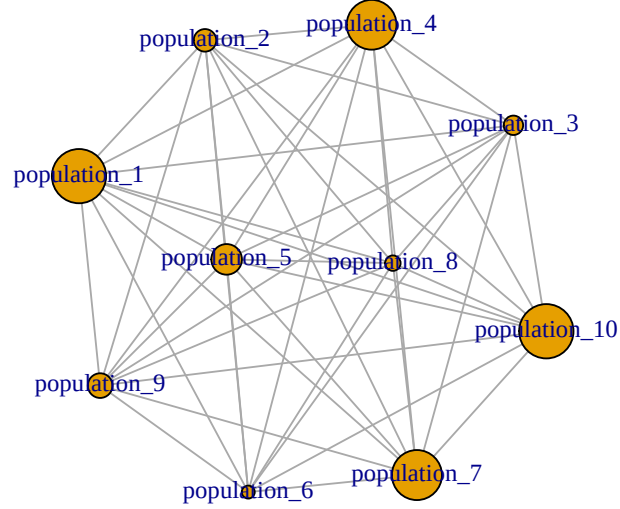
- generate a simple network with 10 vertices fully connected.
- assign population sizes of animals to each vertex
- migration is possible through the network
- migration is dependent on population size
- add edge and vertex characteristics
 - Vertex can be vegetated, have food etc, have human habitaion
 - Edges can have weights associated with them due to roadways, forested networks, rivers/mountains, etc

Fully Connected Migration Network



Vertex characteristics

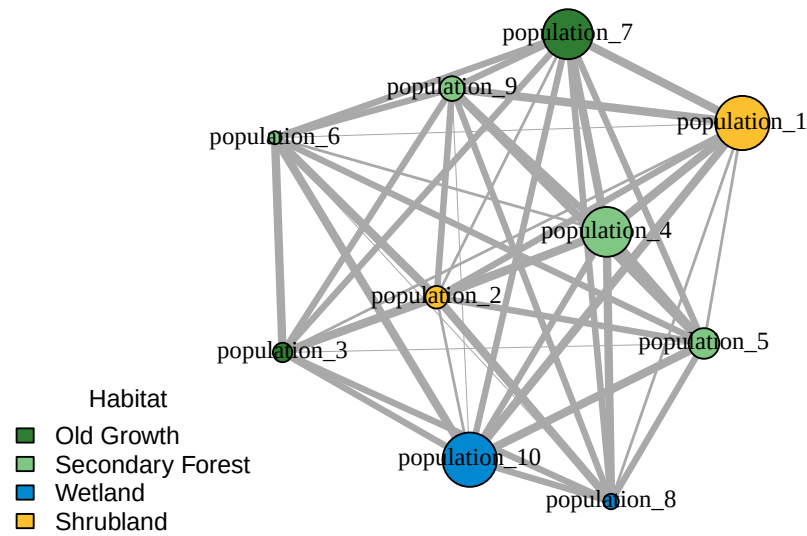
We assign population sizes p_i at each vertex i , with habitat characteristics such as habitat type t_i , human presence (hunting, conflict) h_i , food availability f_i , climate c_i , etc. p_i is evolving with birth, death (due to carrying capacity K , life time), and migration. If $p_i = 0$ for a long time, make $K = 0$.



Edge characteristics

We assign corridor types to edges, which can be wildlife corridor (for example - dense forest, open woodland, etc), terrain (river crossing, mountains, desert plateau), or urbanization (highways, human settlements). Each corridor type has a weight associated with it that affects migration.

Friction maps available for terrain. That can influence edge weight.

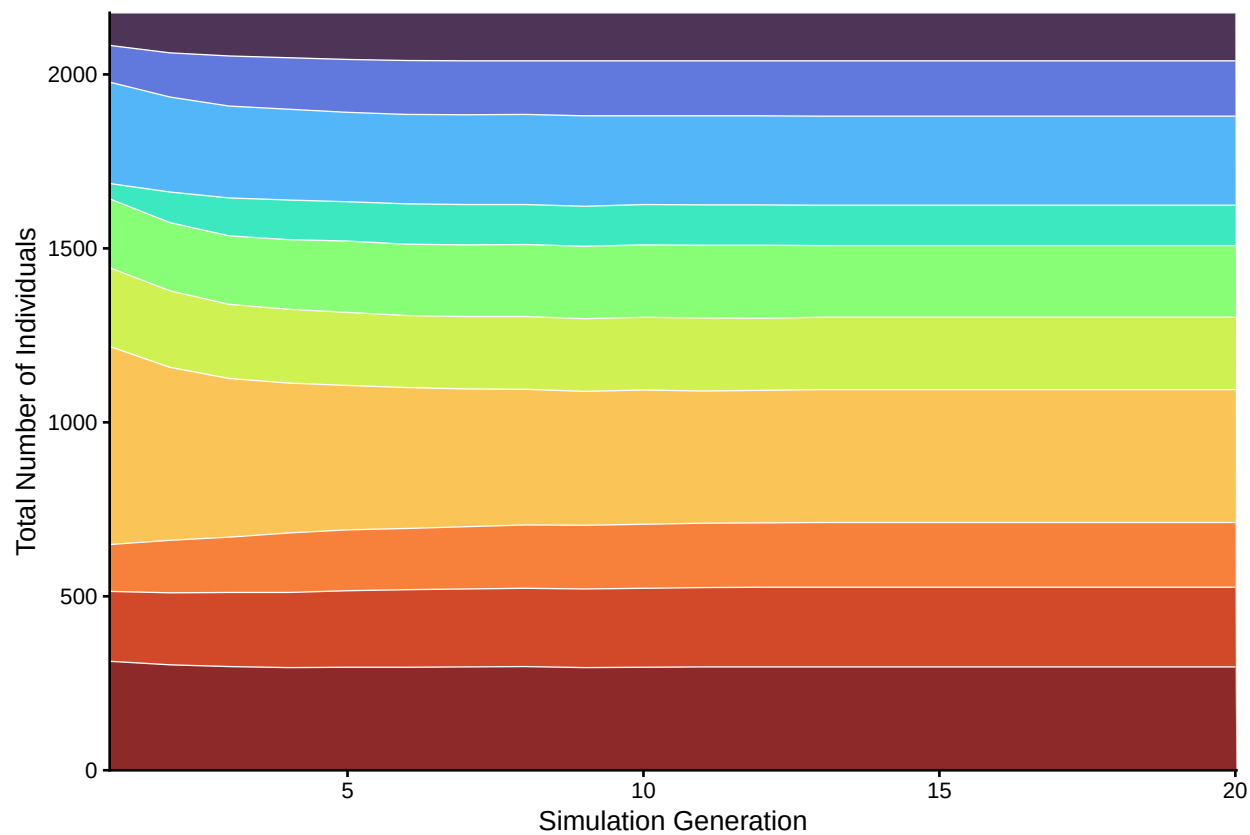


Assign carrying capacity

Based on habitat type, food availability, and human presence.

```
##           name    habitat food_availability human_presence capacity
## population_1 population_1 Shrubland         Medium      Medium      94
## population_2 population_2 Shrubland          High      Medium     105
## population_3 population_3 Old Growth         None       None     300
##           pop_size
## population_1      94
## population_2     106
## population_3     291
## [ reached 'max' / getOption("max.print") -- omitted 7 rows ]
```

Migration, Birth, Death



Final Network State

